

# Patrick Tierney

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## EXPERIENCE

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### Microsoft

**Software Engineer II** (2019 – 2025), **Software Engineer I** (2017 – 2019)

#### Platform & Framework Development

- Architected and led delivery of a testing platform used by 150+ engineers supporting Copilot Hub, standardizing automated testing
- Co-designed and implemented a telemetry platform for Microsoft Whiteboard using OpenTelemetry, making the service easier to diagnose in production while keeping developer overhead low
- Consolidated disparate AI tooling from across Microsoft teams into a unified, maintainable platform codebase
- Defined standards and requirements for an AI-driven static analysis system enforcing testing and compliance
- Enhanced a telemetry query DSL used by 1,000+ Edge engineers, improving analytics and diagnostic capabilities
- Drove migration of Whiteboard's layout engine from C# to TypeScript/React within its backing framework
- Built layout-engine tooling to position visual elements correctly on screen, supporting 80+ engineers as Whiteboard scaled to tens of millions of users
- Owned layout logic for on-screen placement of non-hierarchical graphical elements across 50+ internal applications

#### Operations, Reliability, & Production Support

- Established on-call and incident response practices, reducing incident resolution time by 80% and engineering support workload by 70%
- Maintained 99.9% availability for Microsoft Whiteboard supporting school systems globally during COVID-19
- Built and managed CI/CD for Whiteboard and Copilot Hub, automating builds, testing, and validation
- Designed support and escalation model for sovereign government cloud environments (GCC, GCCH, DoD)
- Built tooling and operational runbooks enabling customer support to resolve issues without engineer escalation
- Developed automation that generated dashboards from telemetry data for new feature rollouts
- Created and maintained Azure DevOps repositories for internal analytics utilities, including CI/CD
- Mentored three interns who all returned to Microsoft as full-time engineers

#### Backend Services & APIs

- Developed Azure microservices enabling secure credential and certificate management for internal infrastructure
- Designed data models and API contracts for programmatic access to Whiteboard resources via Microsoft Graph
- Implemented OAuth-based authentication and JWT access token validation for Whiteboard backend services, integrating with Azure AD scopes and permissions

## TECHNICAL SKILLS

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**Languages & Frameworks:** TypeScript, JavaScript, Node.js, C#, .NET, SQL, React, C++, Python, R

**Infrastructure & Automation:** Azure services, Azure DevOps, YAML-based CI/CD, PowerShell

**Testing, Observability & Security:** Jest, Playwright, OpenTelemetry, Power BI, OAuth/JWT authentication, Distributed Tracing & Observability

## EDUCATION

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**Harvey Mudd College**

Bachelor's Degree, Computer Science and Mathematics; Concentration in Economics

Claremont, CA

May 2016